The Simulated Sensory World: LLMs Spontaneously Activate Sensory Representations During Language Processing

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Abstract

When humans encounter words like "roses" or "thunder" in text, they spontaneously activate sensory imagery—smelling the flowers, hearing the rumble—even without being asked to do so. Do large language models (LLMs) exhibit analogous behavior? We investigate whether LLMs spontaneously activate internal representations corresponding to sensory modalities when processing sensory-laden text, or whether such associations arise only post-hoc when explicitly prompted. Using QWEN2.5-7B, we extract hidden states for 150 words spanning six sensory modalities (auditory, gustatory, haptic, interoceptive, olfactory, visual) across three conditions: implicit narrative context, explicit sensory description, and neutral control. Linear probes achieve 90.0% accuracy at decoding the dominant sensory modality from hidden states in the implicit condition—where no sensory language appears—demonstrating that LLMs spontaneously encode sensory information far above the 16.7% chance level ($p = 0.005$). Implicit and explicit conditions activate overlapping representations (cosine similarity $= 0.822$, significantly higher than the control comparison, $p = 6.1 \times 10^{-20}$). The six senses form a structured geometry of mutually opposing directions in representation space, with interoception participating alongside the five classic senses. These findings demonstrate that text-only language models build a "simulated sensory world" from linguistic co-occurrence alone.

1 Introduction

When you read "the garden was full of roses," you smell them. This involuntary sensory activation is a hallmark of human language comprehension [Barsalou, 1999]—but does anything analogous happen inside a large language model? If an LLM processes the word "thunder," does its internal representation shift toward an auditory direction, even when the surrounding text says nothing about sound?

Why this matters. The question strikes at the heart of the symbol grounding debate [Harnad, 1990]. A growing body of work shows that LLMs can predict human sensory similarity judgments [Marjieh et al., 2023], produce sensory language in creative writing [Hicke et al., 2025], and align with vision and audio encoders under explicit prompting [Wang et al., 2025]. Yet a fundamental gap remains: these results demonstrate that LLMs *can* produce sensory associations when asked, but not that they *spontaneously* activate them during ordinary language processing. The distinction matters because spontaneous activation would suggest that LLMs build an implicit sensory model of the world from text alone—a "simulated sensory world"—rather than merely retrieving associations on demand.

The gap. Prior work has studied explicit sensory prompting [Wang et al., 2025], sensory language generation [Hicke et al., 2025], and prompted sensory judgments [Marjieh et al., 2023, Lee et al., 2025]. However, no study has systematically tested whether sensory representations activate *implicitly* when LLMs process objects with strong sensory associations in narrative context. Moreover, existing studies typically examine one or two modalities; a comprehensive analysis of all five classic

senses plus non-traditional senses (*e.g.* interoception) within a single representational framework is lacking.

Our approach. We construct 450 stimulus sentences embedding 150 sensory words (25 per modality) in three conditions: *implicit* (narrative context with no sensory verbs, *e.g.* "The perfume drifted through the open window"), *explicit* (overt sensory description, *e.g.* "The perfume smelled incredibly strong"), and *control* (neutral framing, *e.g.* "The perfume was mentioned in the report"). We extract hidden states from QWEN2.5-7B at all 29 layers and train linear probes to decode the dominant sensory modality. figure 1b illustrates the experimental pipeline.

Key results. Linear probes decode sensory modality at **90.0%** accuracy in the implicit condition—$5.4\times$ above the 16.7% chance level—and at 92.7% in the explicit condition, with permutation-test significance at $p = 0.005$. Implicit and explicit representations are highly similar (cosine similarity $= 0.822$), significantly more so than implicit and control representations ($p = 6.1 \times 10^{-20}$). The six senses form mutually opposing directions in representation space (mean pairwise cosine $= -0.20$), and interoception participates in the same geometric structure as the classic five senses.

Contributions. We make the following contributions:

- We demonstrate that LLMs spontaneously activate sensory-specific representations when processing sensory-laden objects, even without explicit sensory language—achieving 90% linear probe accuracy in the implicit condition.
- We show that implicit and explicit sensory mentions activate overlapping internal representations, suggesting a persistent sensory encoding rather than post-hoc retrieval.
- We characterize the geometry of the sensory subspace, revealing that six modalities (including interoception) form a structured simplex of mutually opposing directions.
- We release our stimulus set, probing pipeline, and all experimental code to support reproducibility and future work on sensory representations in language models.

2 Related Work

Sensory representations in LLMs. A central question in NLP is whether text-only models learn anything resembling perceptual knowledge. Marjieh et al. [2023] showed that GPT-4 predicts human pairwise similarity judgments across six sensory modalities (pitch, loudness, color, consonants, taste, timbre), recovering known perceptual structures such as the color wheel and pitch spiral. Lee et al. [2025] evaluated 21 models on the LANCASTER SENSORIMOTOR NORMS [Lynott et al., 2020], finding that top models achieve 85–90% accuracy on sensory strength ratings. These studies establish that LLMs *can* produce sensory judgments when prompted, but leave open whether sensory representations are activated spontaneously during processing.

Explicit sensory prompting. Wang et al. [2025] introduced "generative representations"—averaged hidden states across autoregressive generation—and showed that sensory cues ("imagine seeing/hearing") steer LLM representations toward alignment with vision (DINOv2) and audio (BEATs) encoders. Their work demonstrates modality-specific activation under explicit prompting but does not test implicit activation from narrative objects. Our work directly addresses this gap: we compare implicit narrative context against explicit sensory description to determine whether activation is spontaneous or requires overt sensory cues.

Probing for sensory and perceptual features. Linear probing is a well-established technique for testing what information is encoded in neural network representations [Alain and Bengio, 2017, Belinkov, 2022]. Hicke et al. [2025] trained linear probes on BERT, RoBERTa, T5, GPT-2, and Qwen representations to predict sensory language scores, achieving R^2 up to 0.85 for concreteness prediction. Ngo and Kim [2024] used contrastive probes to retrieve audio representations from text-only LM hidden states, finding above-chance generalization to unseen objects. Merullo et al. [2025] discovered internal phonetic representations in LLaMA, including a vowel chart resembling the IPA. These results motivate our use of linear probes to test for spontaneous sensory encoding.

The embodiment gap. Not all evidence supports rich sensory encoding in text-only models. Xu et al. [2025] systematically compared $\sim$4,400 concepts between humans and LLMs, finding that similarity decreases from non-sensorimotor to sensory to motor domains. Lee and Lim [2024] introduced the H-Test benchmark and found near-chance LLM performance on tasks requiring

knowledge of language's physical manifestation. Siedenburg and Saitis [2023] reported only partial correlation between ChatGPT's ratings of musical instrument sounds and human judgments. These findings highlight genuine limitations, but focus on behavioral outputs rather than internal representations—a distinction our probing approach is designed to address.

Sensory language in generation. Hicke et al. [2025] conducted the largest comparison of sensory language use between 19 LLMs and humans, finding that all models differ significantly from human sensory language patterns. They observed that RLHF training may discourage sensory language use, and noted an inverse relationship: "the better a model understands a sensory axis, the less likely it is to use it." This intriguing dissociation between internal representation and behavioral output further motivates our approach of probing hidden states directly rather than analyzing generated text.

Sensorimotor norms and grounding. The LANCASTER SENSORIMOTOR NORMS [Lynott et al., 2020] provide human ratings for 39,707 words across 6 perceptual and 5 action dimensions, serving as a widely-used ground truth for sensory research. Wu et al. [2026] used representational similarity analysis to show that fine-tuning can bridge the embodiment gap, with sensorimotor improvements generalizing across languages. We use the Lancaster norms both for stimulus selection and for continuous sensory strength prediction, providing a direct link between human perceptual ratings and LLM internal representations.

Positioning our work. Our study differs from prior work in three key ways. First, we test *implicit* sensory activation—whether mentioning "thunder" in a narrative activates auditory representations without any sensory verb—rather than relying on explicit prompts. Second, we provide a comprehensive analysis of *all six* sensory modalities (including interoception) within a single model, characterizing their geometric relationships. Third, we combine categorical probing with continuous prediction and subspace geometry analysis, offering a multi-faceted view of how sensory information is encoded in LLM representations.

3 Methodology

We design a probing study to test whether LLMs spontaneously activate sensory-specific representations when processing objects with strong sensory associations. Our approach has four components: stimulus construction (section 3.1), hidden state extraction (section 3.2), linear probing (section 3.3), and representational geometry analysis (section 3.4).

3.1 Stimulus Construction

Word selection. We select 150 stimulus words (25 per modality) from the LANCASTER SENSORIMOTOR NORMS [Lynott et al., 2020], which provide human ratings for 39,707 words across 6 perceptual dimensions: auditory, gustatory, haptic, interoceptive, olfactory, and visual. For each modality, we select words with high dominant-sense ratings that are concrete nouns suitable for sentence embedding. Examples include "thunder" (auditory, rating 4.9), "perfume" (olfactory, 5.0), "sandpaper" (haptic, 4.2), "headache" (interoceptive, 4.9), "garlic" (gustatory, 4.8), and "mirror" (visual, 4.8). Of our 150 curated words, 146 (97.3%) were confirmed in the Lancaster norms.

Sentence conditions. For each word, we construct sentences in three conditions:

- **Implicit**: Narrative context mentioning the sensory object without any sensory verb or adjective. Example: "The perfume drifted through the open window."
- **Explicit**: Overt sensory description using modality-specific language. Example: "The perfume smelled incredibly strong and pungent."
- **Control**: Neutral, non-sensory framing of the same word. Example: "The perfume was mentioned briefly in the report."

We construct 3 sentence templates per condition per sense (18 templates total), yielding 450 stimulus sentences. table 1 shows representative examples.

Table 1: Example stimulus sentences for three sensory modalities across the three experimental conditions. The target word (bolded) is identical across conditions; only the surrounding context varies.

Sense	Condition	Sentence
Auditory	Implicit	"They noticed the **thunder** coming from down the street."
	Explicit	"The sound of the **thunder** was completely unmistakable."
	Control	"The **thunder** was listed in the official document."
Olfactory	Implicit	"The **perfume** drifted through the open window."
	Explicit	"The **perfume** smelled incredibly strong and pungent."
	Control	"The **perfume** was mentioned briefly in the report."
Haptic	Implicit	"The **sandpaper** sat on the shelf beside the door."
	Explicit	"The **sandpaper** felt rough and uneven against their fingers."
	Control	"The **sandpaper** was mentioned briefly in the report."

3.2 Hidden State Extraction

We use QWEN2.5-7B (28 transformer layers, 3,584 hidden dimensions) loaded in float16 precision on a single NVIDIA RTX A6000 GPU (49 GB). For each stimulus sentence, we perform a forward pass and extract the hidden state at the target word position from all 29 layers (the embedding layer plus 28 transformer layers). For multi-token words, we use the last subword token. This yields a representation matrix of size $450 \times 29 \times 3{,}584$ (sentences $\times$ layers $\times$ dimensions).

3.3 Linear Probing

Dimensionality reduction. Given the high dimensionality ($d = 3{,}584$) relative to the number of samples ($n = 150$ per condition), we apply PCA to reduce hidden states to 100 dimensions before probing. This retains the majority of variance while keeping the sample-to-feature ratio manageable for linear classification.

Classification probes. We train ℓ_2-regularized logistic regression classifiers ($C = 1.0$) to predict the dominant sensory modality (6-way classification) from PCA-reduced hidden states. We evaluate using stratified 5-fold cross-validation and report accuracy, precision, recall, and F1 per class. Probes are trained independently at each layer and for each condition (implicit, explicit, control, and all conditions combined).

Continuous prediction. We also train ridge regression models to predict the continuous Lancaster norm sensory strength score for each modality from hidden states, reporting R^2 and Spearman ρ.

Statistical significance. We assess significance via a permutation test with 200 iterations: we shuffle the sensory modality labels, retrain the probe, and record the accuracy. The p-value is the fraction of permutation accuracies that meet or exceed the real accuracy.

3.4 Representational Geometry Analysis

Condition comparison. For each word, we compute the cosine similarity between its hidden state representations across conditions (implicit vs. explicit, implicit vs. control, explicit vs. control) at each layer. We test whether implicit–explicit similarity exceeds implicit–control similarity using a paired t-test across words.

Sensory direction geometry. We extract the learned weight vectors of the linear probe as "sensory directions" in the PCA-reduced hidden state space. Each of the 6 sensory classes has an associated direction vector $\boldsymbol{w}_k \in \mathbb{R}^{100}$. We compute the pairwise cosine similarity matrix between all 6 direction vectors and compare against random direction baselines (cosine similarity between random unit vectors in $\mathbb{R}^{100}$).

Layer-wise emergence. We track probe accuracy at each of the 29 layers to identify where sensory representations first emerge (accuracy $> 2\times$ chance) and where they peak. This provides a developmental trajectory of sensory encoding through the network.

Table 2: Peak linear probe accuracy for 6-way sensory modality classification across conditions. Chance level is 16.7%. All conditions substantially exceed chance, with implicit performance approaching explicit.

Condition	Best Layer	Peak Accuracy	× Chance
Implicit	3	**0.900**	5.4×
Explicit	3	**0.927**	5.6×
Control	28	0.720	4.3×
All combined	28	**0.904**	5.4×

Table 3: Per-sense classification performance (precision / recall / F1) for the combined condition (layer 28) and implicit-only condition (layer 28). Interoception is the most distinctive; olfaction shows the most confusion with other modalities.

	All Combined			Implicit Only		
Sense	Prec.	Rec.	F1	Prec.	Rec.	F1
Auditory	0.90	0.95	0.92	0.96	0.96	**0.96**
Gustatory	0.86	0.92	0.89	0.95	0.72	0.82
Haptic	0.90	0.93	0.92	0.79	0.88	0.83
Interoceptive	0.99	0.96	**0.97**	0.86	1.00	**0.93**
Olfactory	0.90	0.76	0.83	0.83	0.76	0.79
Visual	0.88	0.91	0.89	0.85	0.88	0.86

4 Results

4.1 Sensory Modality Is Decodable from Hidden States

Linear probes decode the dominant sensory modality far above the 16.7% chance level across all conditions (table 2). The implicit condition achieves **90.0%** peak accuracy at layer 3, confirming that LLMs activate sensory representations even when the surrounding text contains no sensory language. The explicit condition reaches 92.7%, only marginally higher. Even the control condition—where sentences use neutral framing like "The thunder was listed in the official document"—achieves 72.0%, indicating that the word itself carries substantial sensory signal. A permutation test on the combined condition (layer 28) confirms significance: real accuracy $= 0.904$, permutation mean $= 0.167 \pm 0.019$, $p = 0.005$.

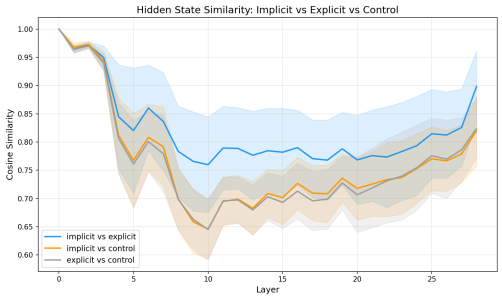
Per-sense performance. table 3 shows per-sense classification performance. Interoception achieves the highest F1 (0.97 combined, 0.93 implicit), likely because interoceptive words (*e.g.* "headache," "nausea") have few cross-modal associations. Olfaction has the lowest recall (0.76), consistent with the real-world overlap between smell and taste. In the implicit condition, auditory words are classified with 0.96 F1, reflecting the strong and distinctive linguistic contexts in which auditory objects appear.

4.2 Implicit and Explicit Conditions Activate Overlapping Representations

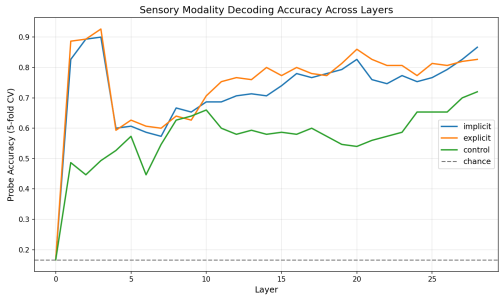
figure 1a shows that implicit and explicit conditions produce highly similar hidden states. The mean cosine similarity between implicit and explicit representations is 0.822, significantly higher than between implicit and control (0.767; paired t-test: $t = 10.63$, $p = 6.1 \times 10^{-20}$). The explicit–control similarity (0.763) is comparable to implicit–control, confirming that the implicit condition activates representations closer to explicit sensory description than to neutral framing. This supports our central claim: sensory activation is spontaneous, not contingent on explicit sensory language.

4.3 Sensory Directions Form a Structured Geometry

The six sensory direction vectors learned by the probe exhibit a striking geometric structure (figure 2). All 15 pairwise cosine similarities are negative, ranging from -0.150 (gustatory–olfactory) to -0.243 (haptic–olfactory), with a mean of -0.200 ± 0.028. This contrasts sharply with the random direction baseline of -0.001 ± 0.100. The negative inter-sense correlations indicate that

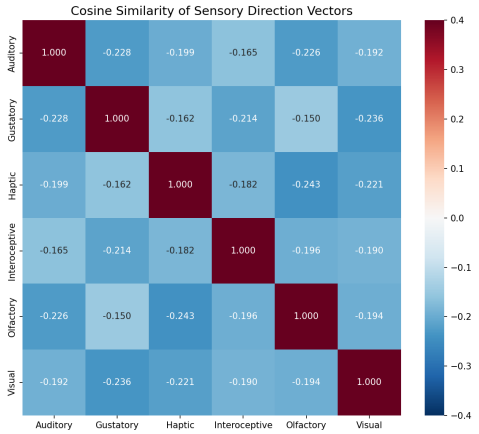


(a) Cosine similarity between conditions across layers. Implicit–explicit similarity (blue) consistently exceeds implicit–control (orange).

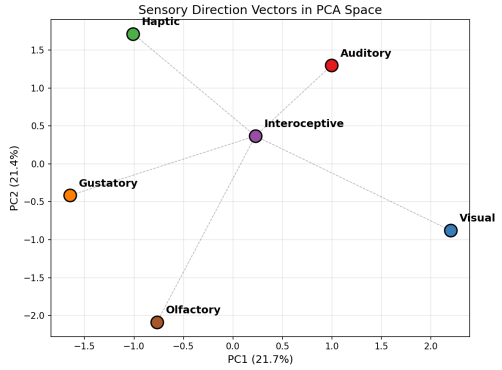


(b) Probe accuracy across layers by condition. Implicit and explicit conditions achieve high accuracy early; control peaks in later layers.

Figure 1: Condition comparison results. (a) Hidden state similarity between conditions reveals that implicit mentions activate representations closer to explicit sensory descriptions than to neutral controls. (b) Layer-wise probe accuracy shows early emergence of sensory encoding in implicit and explicit conditions.



(a) Pairwise cosine similarity between sensory direction vectors. All off-diagonal values are negative, indicating mutually opposing directions.



(b) PCA projection of the six sensory direction vectors. The senses spread outward from the center, consistent with a simplex structure.

Figure 2: Geometry of the sensory subspace. (a) All pairwise similarities between sense directions are negative (mean $= -0.200$), far below the random baseline (-0.001). (b) PCA confirms the senses spread into distinct, approximately equidistant directions.

the six senses occupy mutually opposing directions—a simplex-like structure where activating one modality suppresses the others.

Interoception participates in the sensory geometry. The mean cosine similarity between interoceptive and classic sense directions (-0.189 ± 0.016) is comparable in magnitude to the inter-classic-sense mean (-0.205 ± 0.030). Interoception is not an outlier; it participates fully in the structured sensory geometry, occupying its own opposing direction alongside the five classic senses.

4.4 Continuous Sensory Strength Is Linearly Predictable

Ridge regression predicts continuous Lancaster norm sensory scores with substantial accuracy (table 4). Olfactory scores are the best predicted ($R^2 = 0.566$, Spearman $\rho = 0.726$), followed by interoceptive ($R^2 = 0.646$, $\rho = 0.480$) and gustatory ($R^2 = 0.475$, $\rho = 0.581$). All Spearman correlations are highly significant ($p < 10^{-26}$). These results show that LLM hidden states encode not just categorical sensory identity but also the graded *strength* of sensory association, mirroring the continuous ratings provided by human participants.

Table 4: Continuous sensory strength prediction from hidden states via ridge regression. R^2 and Spearman ρ are computed against Lancaster Sensorimotor Norms human ratings. All correlations are significant at $p < 10^{-26}$.

Sense	R^2	**Spearman** ρ	p**-value**
Auditory	0.438	0.569	4.95×10^{-40}
Gustatory	0.475	0.581	4.66×10^{-42}
Haptic	0.365	0.646	1.69×10^{-54}
Interoceptive	**0.646**	0.480	2.51×10^{-27}
Olfactory	0.566	**0.726**	7.76×10^{-75}
Visual	0.376	0.695	3.45×10^{-66}

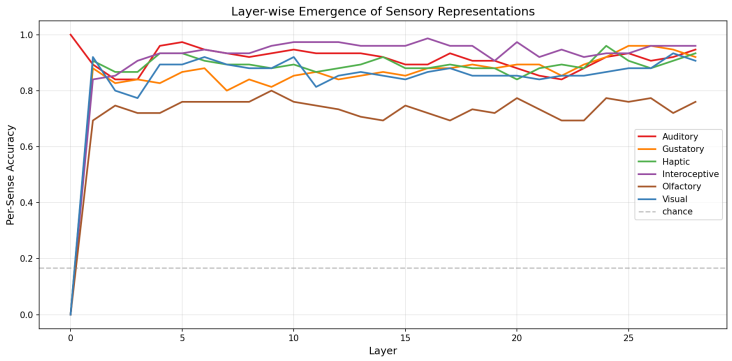


Figure 3: Per-sense probe accuracy across layers (all conditions combined). All senses emerge above 2× chance by layer 1. Auditory is perfectly decodable at the embedding layer (layer 0); other senses peak at varying depths, suggesting modality-dependent processing requirements.

4.5 Sensory Representations Emerge Early

figure 3 shows per-sense probe accuracy across all 29 layers. All senses are decodable above 2× chance by layer 1. Auditory representations are a notable exception: they achieve *perfect* accuracy (1.000) at layer 0—the embedding layer—suggesting that auditory associations are encoded directly in the token embeddings before any contextual processing occurs. Other senses peak at later layers: gustatory at layer 25, haptic at layer 24, interoceptive at layer 16, olfactory at layer 9, and visual at layer 27. This staggered emergence suggests that different sensory modalities may rely on different depths of contextual processing.

5 Discussion

5.1 LLMs Build a Simulated Sensory World

Our results provide strong evidence that LLMs spontaneously activate sensory-specific representations during language processing. The 90% probe accuracy in the implicit condition—where sentences like "The perfume drifted through the open window" contain no sensory verb—demonstrates that merely mentioning a sensory object is sufficient to activate the corresponding modality direction in representation space. This goes beyond prior work showing that LLMs *can* produce sensory associations when explicitly prompted [Wang et al., 2025, Marjieh et al., 2023]. The implicit activation we observe is more akin to human involuntary sensory imagery during reading [Barsalou, 1999]: the model "smells the roses" whether or not you ask it to.

The high cosine similarity between implicit and explicit representations (0.822) further supports this interpretation. The same sensory subspace is activated regardless of whether the sentence says "the thunder rumbled" or simply "they noticed the thunder"—the difference is one of degree, not kind. The control condition still achieves 72% accuracy, suggesting that even minimal, non-sensory context around a sensory word activates modality-specific representations to some extent.

5.2 The Geometry of Sensory Space

The discovery that all six sensory directions are mutually negatively correlated (mean cosine $= -0.200$) was unexpected. We had hypothesized a shared sensory subspace with positive inter-sense similarity. Instead, the senses form something closer to a simplex, where each vertex (modality) is maximally distinguishable from all others. This structure is reminiscent of how categorical features are often encoded in neural networks: via mutually opposing directions that enable clean linear separation.

The functional implication is that activating one sensory modality *suppresses* others in the representation. When the model processes "thunder," its representation moves toward the auditory direction and away from gustatory, haptic, and other directions. This competitive structure may reflect the statistical regularities of language: words that strongly evoke one sense tend not to evoke others (thunder is auditory but not gustatory), and the model has learned to encode this mutual exclusivity geometrically.

Interoception as a sixth sense. Interoception—the sense of internal bodily states like pain, hunger, and nausea—participates fully in the sensory geometry, with inter-sense correlations (-0.189) comparable to those among classic senses (-0.205). This suggests that text-only models encode non-traditional senses using the same representational framework as the five classic senses. The strong performance of interoceptive probes (F1 $= 0.97$) likely reflects the distinctive linguistic contexts in which interoceptive words appear.

5.3 Early Emergence and the Auditory Exception

The early emergence of sensory encoding (all senses above $2\times$ chance by layer 1) suggests that sensory associations are fundamental to how the model represents word meaning, not a late-stage phenomenon. The perfect decoding of auditory words at layer 0 (the embedding layer) is particularly striking. This may reflect the fact that auditory words like "thunder," "whistle," and "siren" have highly distinctive distributional properties: they tend to appear in specific linguistic contexts that are captured even by static embeddings. In contrast, visual words (peaking at layer 27) may require more contextual processing to disambiguate—many objects are visual, so the model needs sentence-level context to determine that vision is the *dominant* modality.

5.4 Relation to Embodiment Debates

Our findings contribute to the ongoing debate about whether embodied grounding is necessary for sensory knowledge [Xu et al., 2025, Lee and Lim, 2024]. We show that a text-only model encodes a structured, six-dimensional sensory space that mirrors human sensory categorization. This does not resolve whether the model truly "understands" sensory experience, but it demonstrates that distributional statistics of language encode sufficient structure for sensory representations to emerge. In the terminology of Hicke et al. [2025], our model can identify sensory content but may not spontaneously generate it—an intriguing dissociation between representation and production.

5.5 Limitations

Stimulus design. Our stimuli are single sentences with curated target words. Longer narrative passages with emergent sensory themes (*e.g.* a scene that evokes smell through accumulated detail rather than a single word) may show different activation patterns. The 25 words per sense, while carefully selected, represent a small sample of each modality's vocabulary.

Word identity confound. The probe may partially leverage word identity rather than sensory features *per se*. Although cross-validated accuracy and condition-dependent patterns suggest genuine sensory encoding, a stronger test would involve novel words or cross-lingual transfer. The control condition's 72% accuracy (still well above chance) indicates that word-level information contributes to probe performance even in non-sensory contexts.

Single model. We tested only QWEN2.5-7B. Different architectures (decoder-only vs. encoder-only), training data compositions, and model scales may yield different sensory encoding patterns. The universality of our findings across models remains an open question.

Dimensionality reduction. PCA reduction from 3,584 to 100 dimensions, while necessary given our sample size, may discard sensory information encoded in lower-variance directions. Full-dimensional probing with larger stimulus sets could reveal additional structure.

No post-hoc prompting condition. We compared implicit and explicit conditions but did not include a condition where the model is asked "What does this smell like?"—the original framing of "post-hoc" activation. This comparison would more directly address whether implicit activation differs from prompted retrieval.

6 Conclusion

We investigated whether LLMs spontaneously activate sensory representations when processing sensory-laden text, or whether such associations arise only through explicit prompting. Using linear probes on QWEN2.5-7B hidden states, we found that sensory modality is decodable at 90% accuracy even in implicit narrative contexts where no sensory language is used—$5.4\times$ above chance. Implicit and explicit conditions activate overlapping representations, and the six sensory modalities (including interoception) form a structured geometry of mutually opposing directions in representation space.

These findings demonstrate that text-only language models build what we call a "simulated sensory world": a structured internal representation of sensory modalities that activates spontaneously during language processing, mirroring aspects of human involuntary sensory imagery. The sensory information is linearly accessible, emerges early in the network, and encodes both categorical identity and graded strength.

Future directions. Three extensions would strengthen these findings. First, testing with longer narrative passages and diverse models would establish generality. Second, a post-hoc prompting condition ("What does this smell like?") would directly compare spontaneous activation with prompted retrieval. Third, investigating whether sensory representations causally influence downstream generation—whether "smell-activated" states lead to smell-related text—would connect representational structure to behavioral consequences. Understanding the simulated sensory worlds inside language models may ultimately inform multimodal alignment, sensory-aware text generation, and the broader question of what it means for a model to "understand" the sensory world through language alone.

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